METRIC

MIL-P-24705/1A(SH)
19 July 1992
SUPERSEDING
MIL-P-24705/1(SH)
7 June 1990

MILITARY SPECIFICATION SHEET

PENETRATORS, MULTIPLE CABLE, FOR ELECTRIC CABLES, RECTANGULAR METAL FRAME, FOR SQUARE-FACED BLOCKS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-P-24705

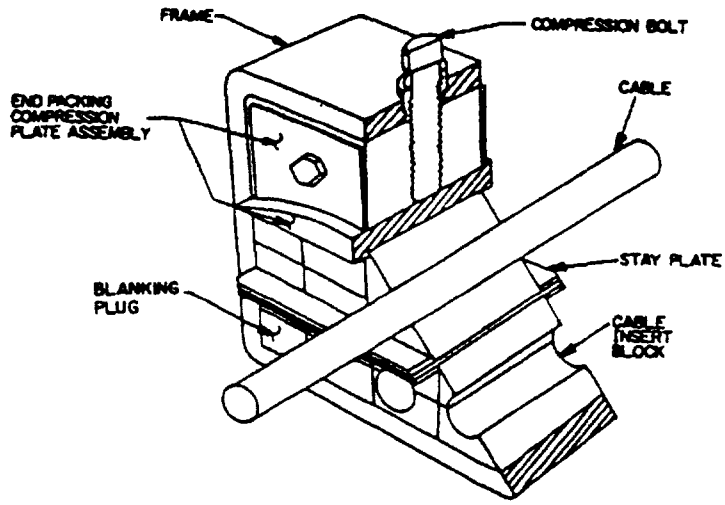


FIGURE 1. *Rectangular multiple cable penetrator - assembled view - typical.*

NOTES

1. Dimensions are in millimeters
2. Inch equivalents are based on 25.4 millimeters equal to one inch and all are given for general information only
3. Unless otherwise specified, tolerances are plus or minus 5 mm (plus or minus .02)
4. Unless otherwise specified, for molded rubber parts, tolerance shall be in accordance with MIL-HDBK-149 (commercial). commercial.
5. All threads shall be in English dimensions No metric threads will be supplied.

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Assembly

Components required for each complete rectangular metal frame multiple cable penetrator assembly specified by this specification sheet are.

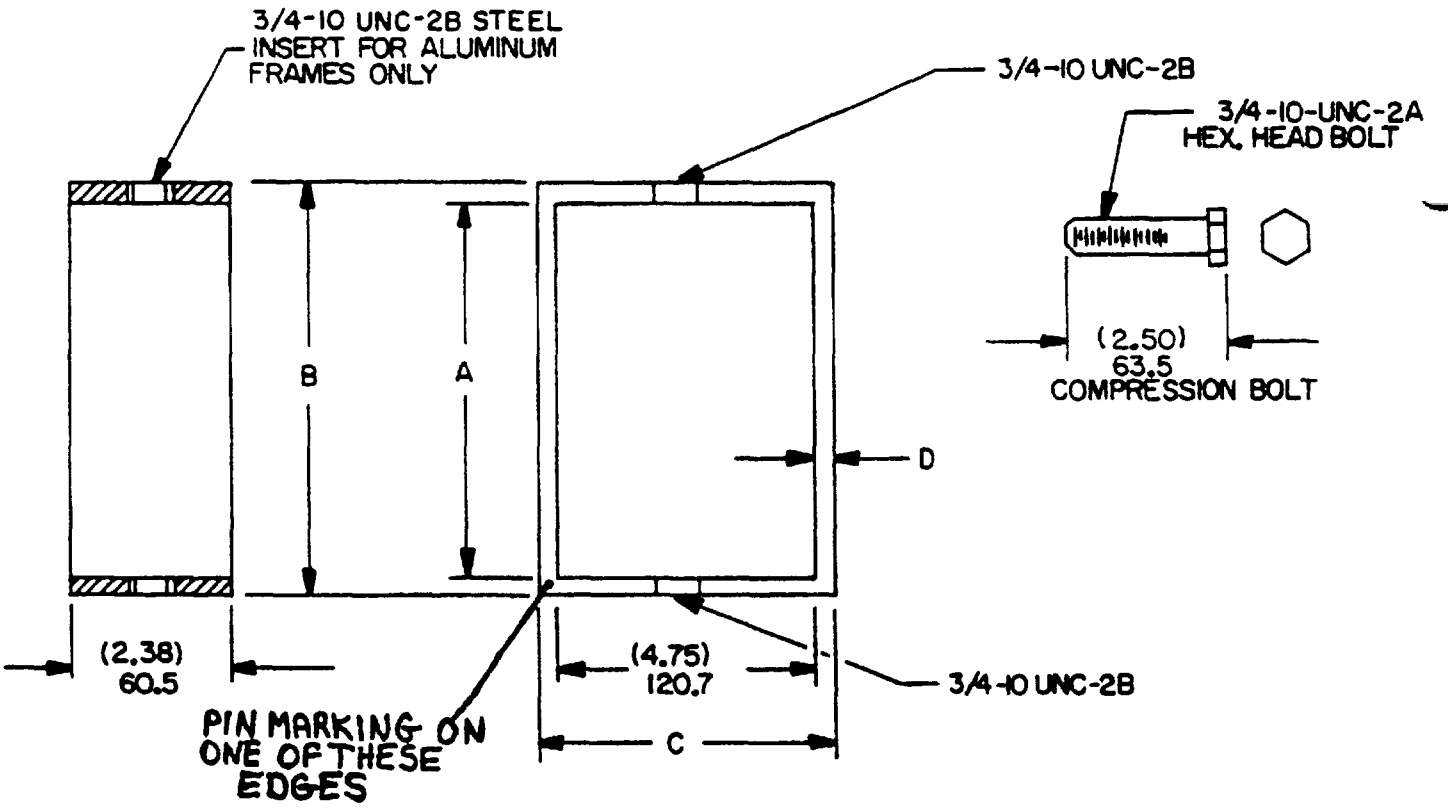
a One rectangular metal frame with associated hardware (see figure 2)

b The appropriate end packing assembly and compression plate and all hardware (see figure 3, 4, 5, 6 and 7)

c Stay plates (see figure 8 or 9 as required, one for each block layer)

d. Square-faced cable insert blocks, blanking insert blocks and filler inserts to fill the interior of the rectangular frame (see figures 10, 11, and 12).

Construction



SH 132316779

FIGURE 2 *Rectangular frame with associated hardware.*

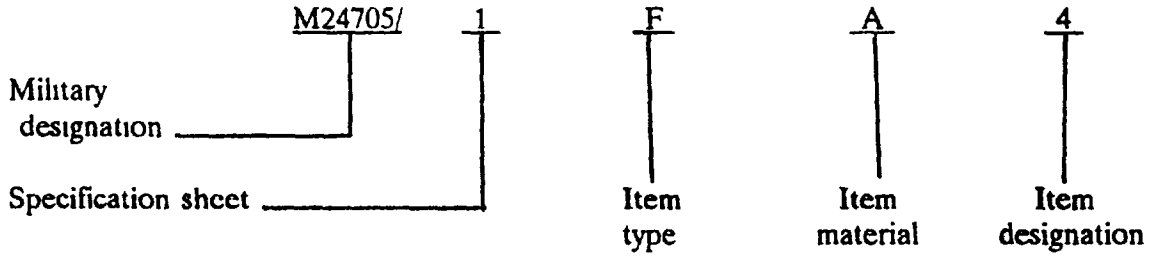
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TABLE I. *Rectangular frame dimensions*

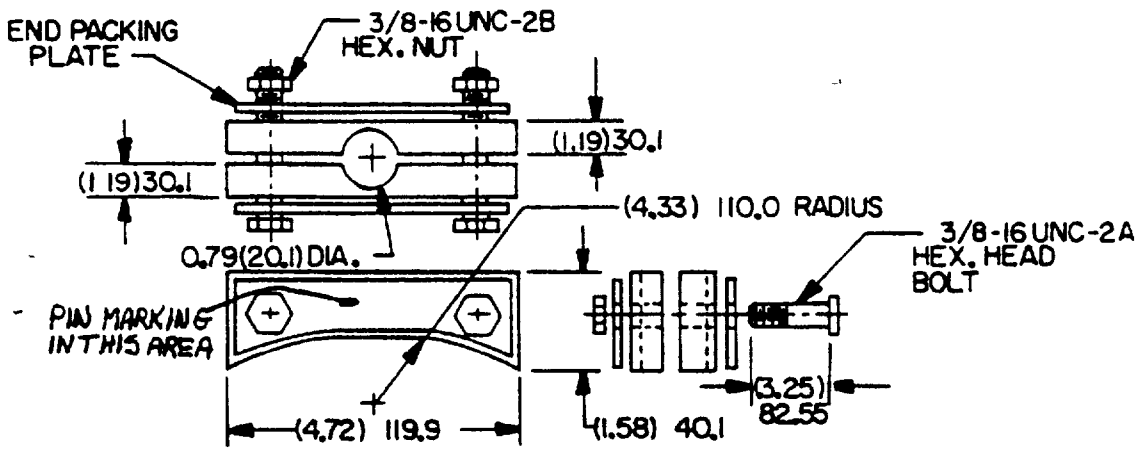
Designator	A	B	C	D
FA2	100.8 (3.97)	126.2 (4.97)	146.3 (5.76)	12.7 (0.50)
FA4	159.5 (6.28)	184.9 (7.28)		
FA6	218.2 (8.59)	243.6 (9.59)		
FA8	277.1 (10.91)	302.5 (11.91)		
FC2, FS2	100.8 (3.97)	119.9 (4.72)	140.0 (5.51)	9.6 (0.38)
FC4, FS4	159.5 (6.28)	178.6 (7.03)		
FC6, FS6	218.2 (8.59)	237.2 (9.34)		
FC8, FS8	277.1 (10.91)	296.2 (11.66)		

PIN (see MIL-P-24705)

EXAMPLE

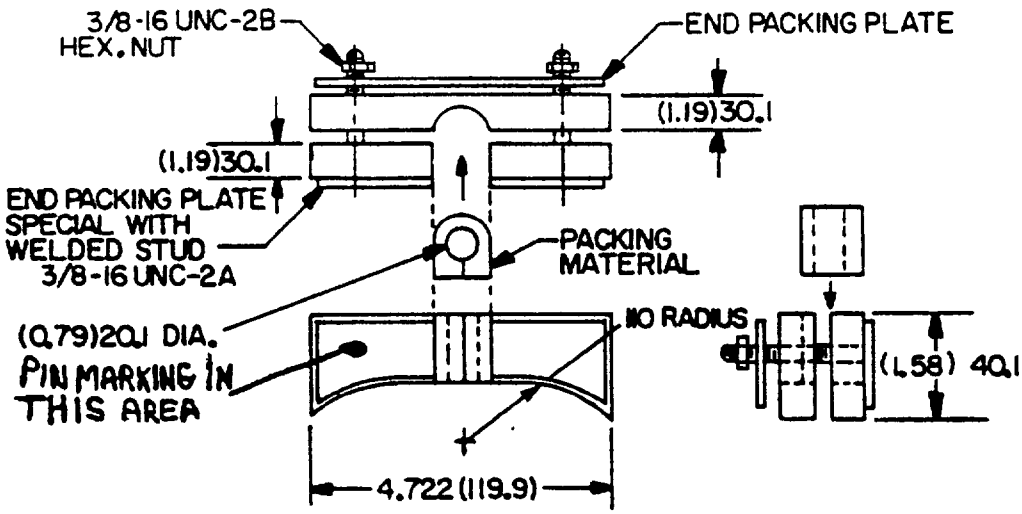


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FIGURE 3. *Style 1A end packing assembly.*



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FIGURE 4 *Style 1B end packing assembly.*

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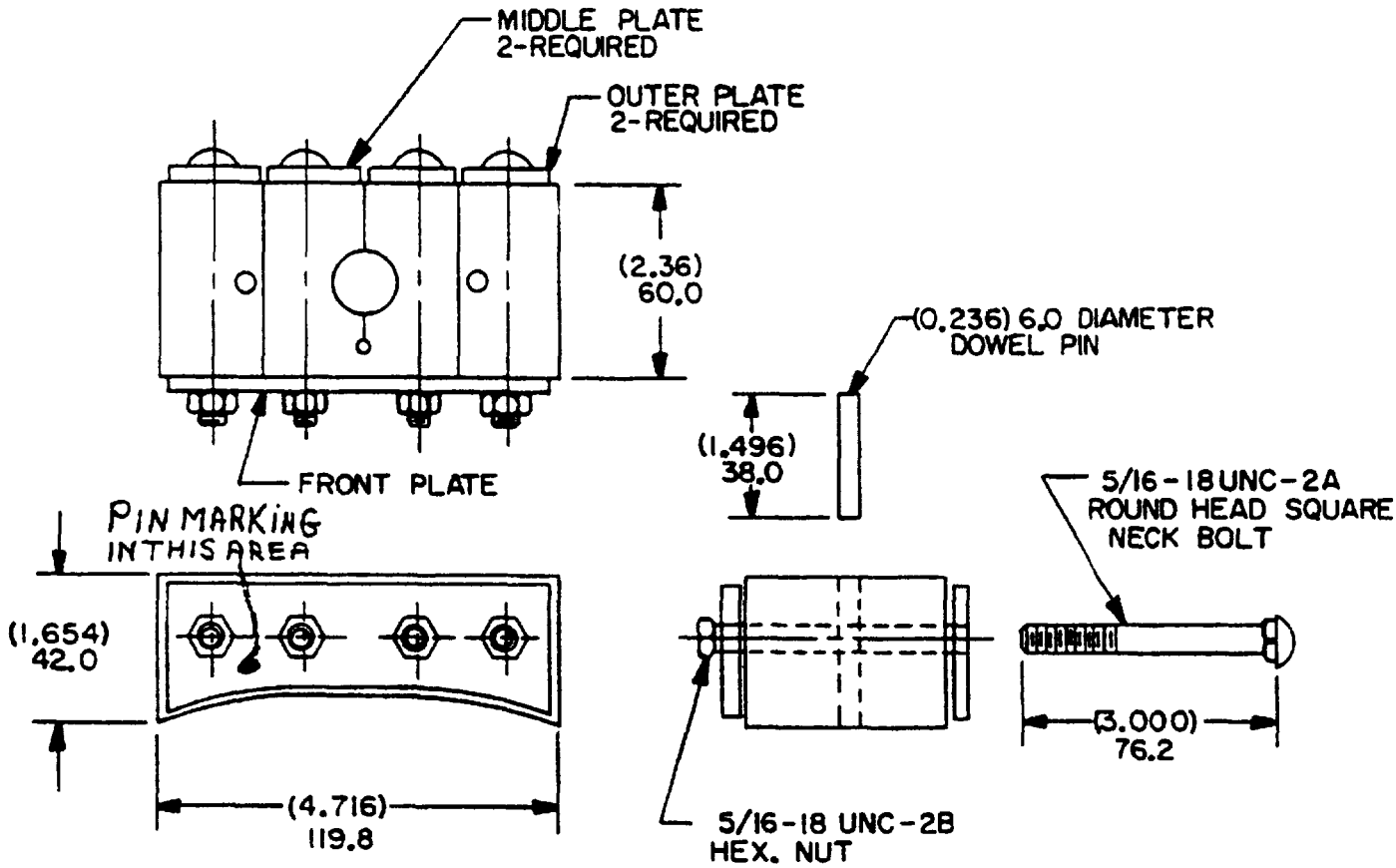
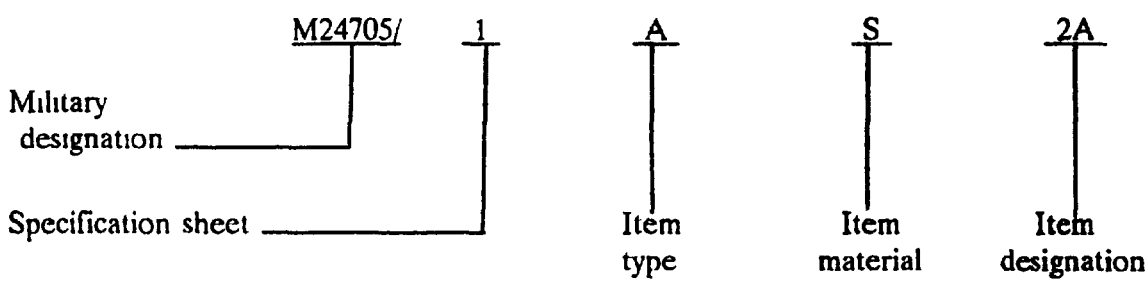


FIGURE 5 *Style 2A end packing assembly.*

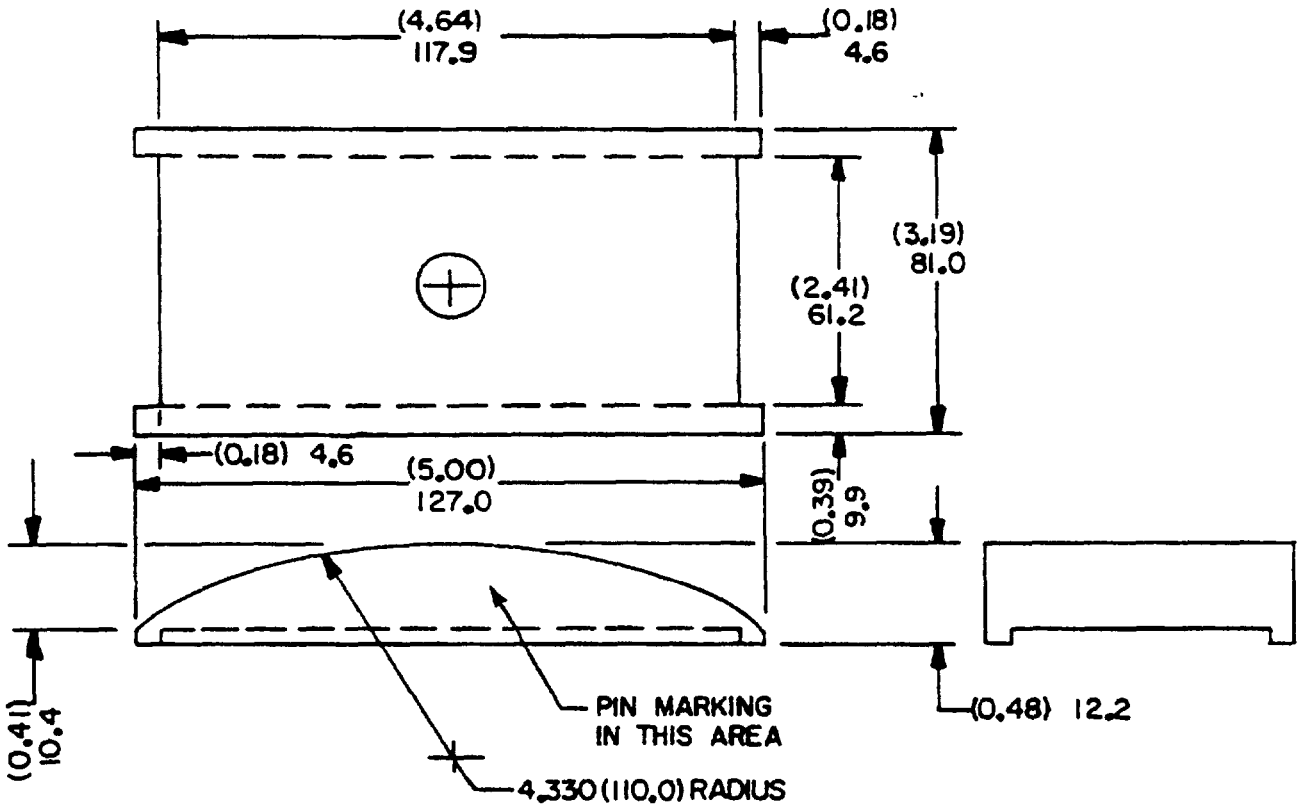
PIN (see MIL-P-24705)

The material specified for end packing assemblies in the military designation pertains to hardware only. The material shall be CRES (C) or steel (S).

EXAMPLE



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NOTE: The compression plate shall be provided with suitable means to restrain movement of the compression plate as compression takes place when assembling

FIGURE 6 *Compression plate for style 1A and 1B end packing.*

Example

	M24705/	1	C	S	1A[1]/
Military Designation			Item type	Item material	Item designation
Specification sheet					

1/ 1A should be used for either 1A or 1B end packing.

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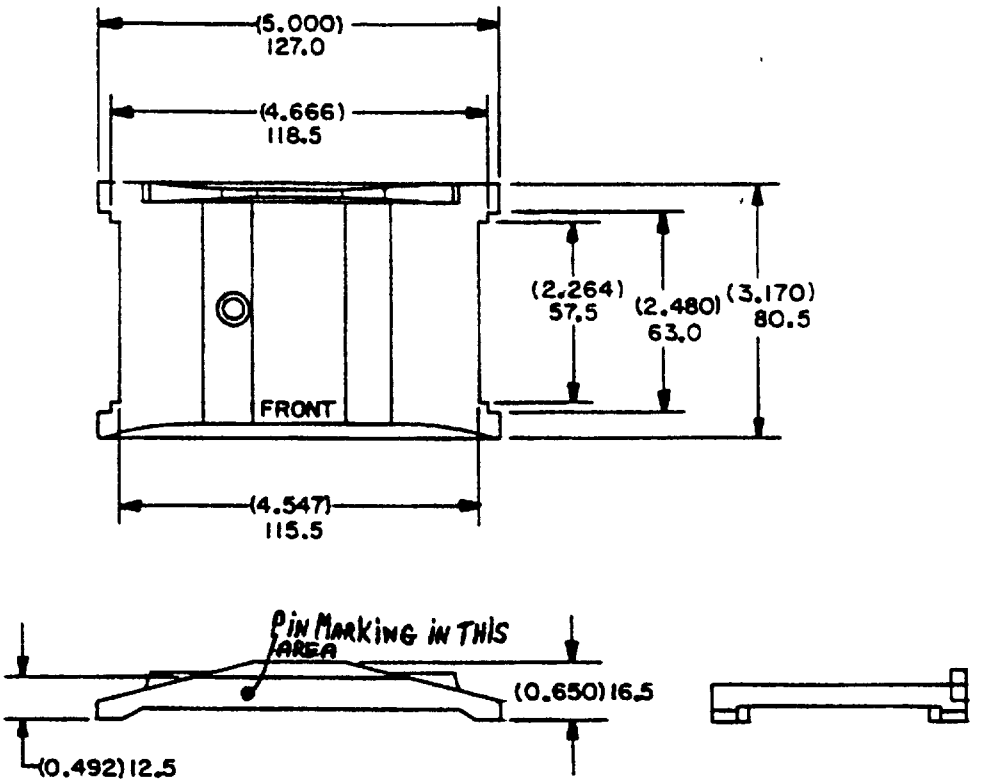


FIGURE 7 Compression plate for style 2A end packing.

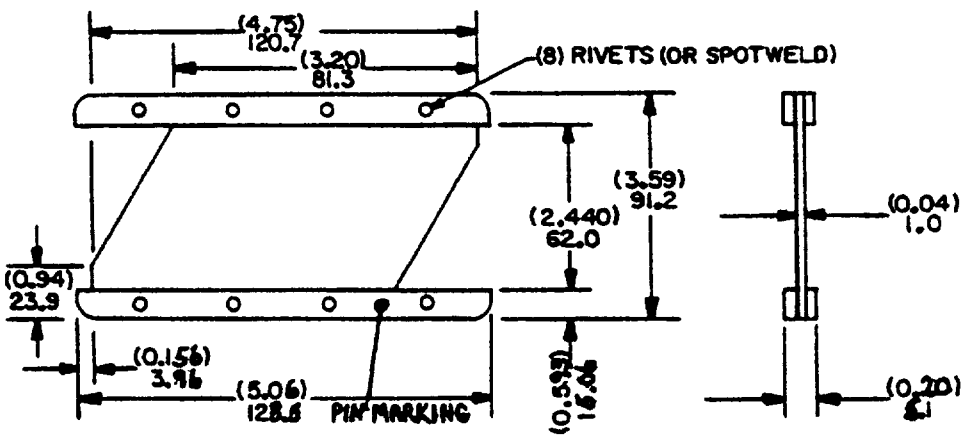


FIGURE 8. *Style 1 stay plate*

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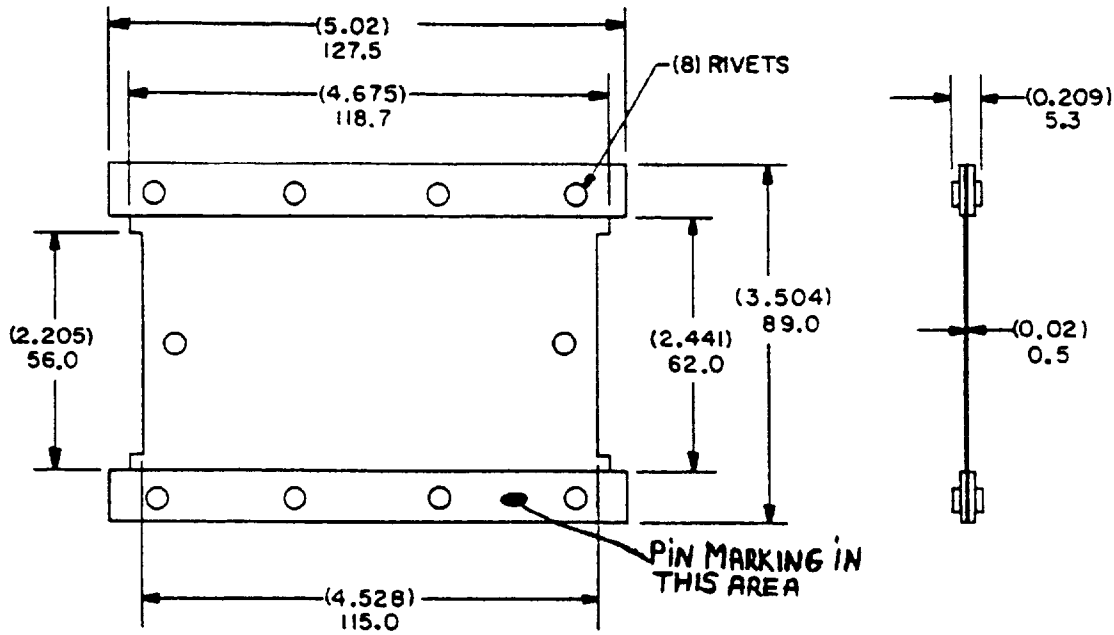
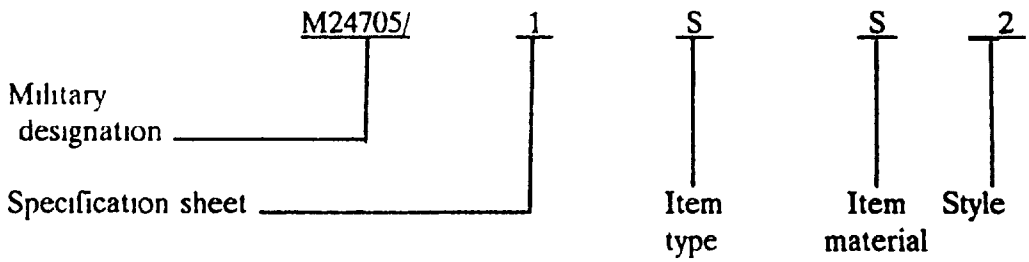


FIGURE 9 Style 2 stay plate

PIN (see MIL-P-24705)

The material specified for stay plates in the military designation shall be aluminum (A), brass (B), CRES (C), or steel (S) The item designation shall consist of a single digit corresponding to a particular style

EXAMPLE



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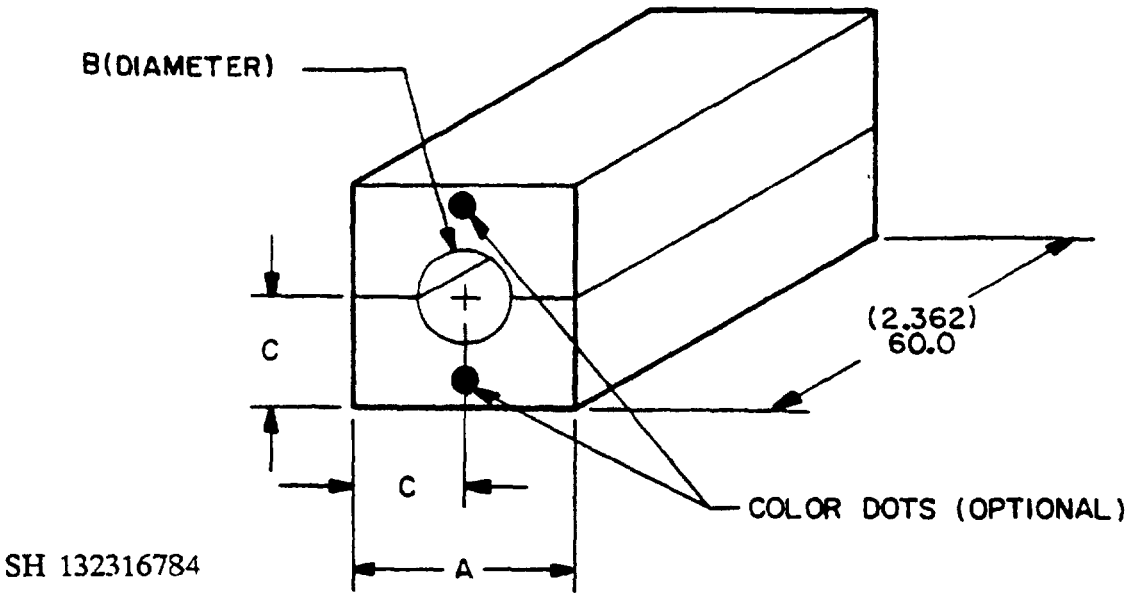


FIGURE 10 *Two-piece square-faced cable insert block.*

TABLE II *Two-piece square-faced cable insert block dimensions*

Designator	A	B	C	Color dots
BN1503/5	15 0 (0 59)	4 02 (0 16)	7.5 (0 30)	Red
BN1504		4 0 (0 15)		
BN1505		5 0 (0 20)		
BN1505/7		6 03 (0 24)		White
BN1506		6 0 (0 24)		
BN1507		7 0 (0 28)		
BN1507/9		8 04 (0 32)		Blue
BN1508		8 0 (0 32)		
BN1509		9.0 (0 35)		
BN2003/6	20 0 (0 79)	4 02 (0 16)	10 0 (0 39)	Green
BN2004		4 0 (0 15)		
BN2005		5 0 (0 20)		
BN2006		6 0 (0 24)		
BN2006/9		7 04 (0.28)		Pink
BN2007		7 0 (0 28)		
BN2008		8 0 (0 32)		
BN2009		9 0 (0 35)		
BN2009/12		10 05 (0 4)		Brown
BN2010		10 0 (0 4)		
BN2011		11 0 (0 43)		
BN2012		12 0 (0 47)		
BN2013		13 0 (0 51)		
BN2013/16		14 07 (0 55)		Gold
BN2014		14 0 (0 55)		
BN2015		15 0 (0 59)		
BN2016		16 0 (0 63)		

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TABLE II. *Two-piece square-faced cable insert block dimensions - Continued.*

Designator	A	B	C	Color dots
BN3012	30.0 (1.18)	12.0 (0.47)	15.0 (0.59)	
BN3012/15		13.07(0.52)		Red
BN3013		13.0 (0.51)		
BN3014		14.0 (0.55)		
BN3015		15.0 (0.59)		
BN3015/18		16.08(0.63)		White
BN3016		16.0 (0.63)		
BN3017		17.0 (0.67)		
BN3018		18.0 (0.71)		
BN3018/21		19.10(0.75)		Blue
BN3019		19.0 (0.75)		
BN3020		20.0 (0.79)		
BN3021		21.0 (0.83)		
BN3021/24		22.11(0.87)		Orange
BN3022		22.0 (0.87)		
BN3023		23.0 (0.91)		
BN3024		24.0 (0.95)		
BN3025		25.0 (0.98)		
BN4022	40.0 (1.58)	22.0 (0.87)	20.0 (0.79)	
BN4022/25		23.12(0.91)		Red
BN4023		23.0 (0.91)		
BN4024		24.0 (0.95)		
BN4025		25.0 (0.98)		
BN4025/28		26.13(1.03)		White
BN4026		26.0 (1.02)		
BN4027		27.0 (1.06)		
BN4028		28.0 (1.10)		
BN4028/31		29.15(1.15)		Blue
BN4029		29.0 (1.14)		
BN4030		30.0 (1.18)		
BN4031		31.0 (1.22)		
BN4031/34		32.16(1.27)		Orange
BN4032		32.0 (1.26)		
BN4033		33.0 (1.30)		
BN4034		34.0 (1.34)		
BN4035		35.0 (1.38)		
BN6031/34	60.0 (2.36)	32.16(1.27)	30.0 (1.18)	Red
BN6032		32.0 (1.26)		
BN6033		33.0 (1.30)		
BN6034		34.0 (1.34)		
BN6034/37		35.18(1.39)		White
BN6035		35.0 (1.38)		
BN6036		36.0 (1.42)		
BN6037		37.0 (1.46)		
BN6037/40		38.19(1.50)		Blue
BN6038		38.0 (1.50)		
BN6039		39.0 (1.54)		
BN6040		40.0 (1.58)		

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TABLE II Two-piece square-faced cable insert block dimensions - Continued

Designator	A	B	C	Color dots
BN6040/43	60.0 (2.36)	41.21 (1.62)	30.0 (1.18)	Orange
BN6041		41.0 (1.61)		
BN6042		42.0 (1.65)		
BN6043		43.0 (1.69)		
BN6043/46		44.22 (1.74)		Purple
BN6044		44.0 (1.74)		
BN6045		45.0 (1.77)		
BN6046		46.0 (1.81)		
BN6046/49		47.24 (1.86)		Yellow
BN6047		47.0 (1.85)		
BN6048		48.0 (1.89)		
BN6049		49.0 (1.93)		
BN6049/52		50.25 (1.98)		Green
BN6050		50.0 (1.97)		
BN6051		51.0 (2.01)		
BN6052		52.0 (2.05)		
BN6053		53.0 (2.09)		
BN6054		54.0 (2.13)		
BN6054/56		54.61 (2.15)		Pink
BN6055		55.0 (2.17)		
BN8054	80.0 (3.15)	54.0 (2.13)	40.0 (1.58)	
BN8056		56.0 (2.21)		
BN8058		58.0 (2.28)		
BN8060		60.0 (2.36)		
BN8062		62.0 (2.44)		
BN8064		64.0 (2.52)		
BN8066		66.0 (2.60)		
BN8068		68.0 (2.68)		
BN8070		70.0 (2.76)		
BN9050	90.0 (3.54)	50.0 (1.97)	45.0 (1.77)	
BN9051		51.0 (2.01)		
BN9052		52.0 (2.05)		
BN9053		53.0 (2.09)		
BN9053/56		54.27 (2.14)		Red
BN9054		54.0 (2.13)		
BN9055		55.0 (2.17)		
BN9056		56.0 (2.21)		
BN9056/59		57.29 (2.26)		White
BN9057		57.0 (2.24)		
BN9058		58.0 (2.28)		
BN9059		59.0 (2.32)		
BN9059/62		60.30 (2.37)		Blue
BN9060		60.0 (2.36)		
BN9061		61.0 (2.40)		
BN9062		62.0 (2.44)		
BN9062/65		63.32 (2.49)		Orange

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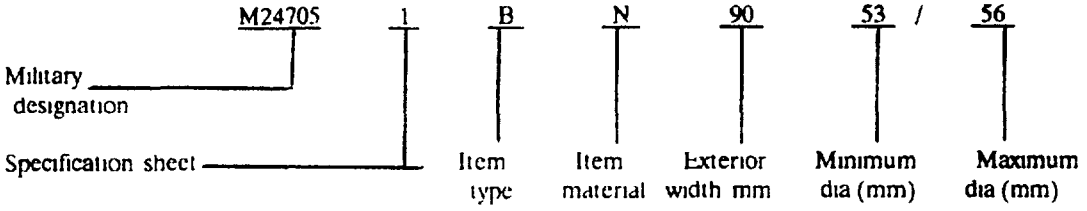
TABLE II Two-piece square-faced cable insert block dimensions - Continued

Designator	A	B	C	Color dots
BN9063	90 0 (3 54)	63 0 (2 48)	45 0 (1 77)	
BN9064		64 0 (2 52)		
BN9065		65 0 (2 56)		
BN9065/68		66 33(2 61)		Purple
BN9066		66 0 (2 60)		
BN9067		67 0 (2 64)		
BN9068		68 0 (2 68)		
BN9068/71		69 35(2 73)		Yellow
BN9069		69 0 (2 72)		
BN9070		70 0 (2 76)		

Military part number (see MIL-P-24705)

The item designation for BN15 thru BN90** square-faced cable insert blocks shall consist of four digits. The first two digits indicate the exterior width of the cable insert block in mm, the last two indicate the passage diameter in mm. The item designation for BN15 thru BN90**/** square-faced cable insert blocks with "COLOR DOTS" shall consist of six digits and a / after the fourth digit. The first two digits indicate the exterior width of the cable insert block in mm, the two digits before the / indicate the minimum diameter in mm, and the last two after the / indicate the maximum diameter in mm.

EXAMPLE



Item designation

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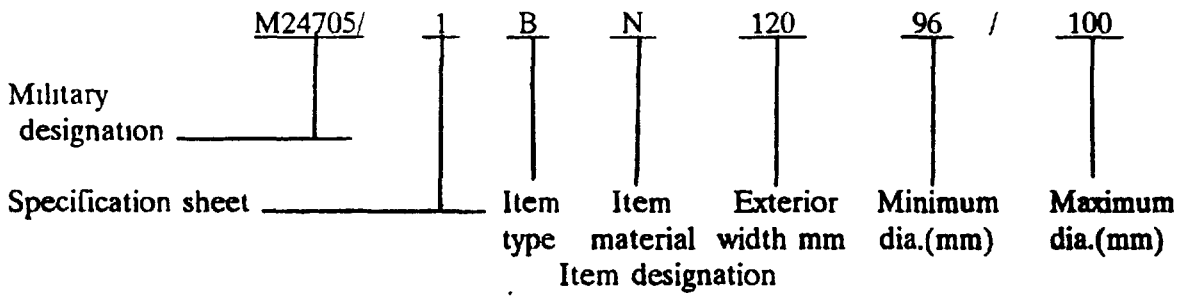
TABLE III Two-piece square-faced cable insert block dimensions.

Designator	A	B	C	Color dots
BN12072/75	120 0 (4 72)	72 3 (2 85)	60.0 (2.36)	Red
BN12075		75 0 (2 95)		
BN12075/78		76 0 (2 99)		White
BN12078/81		79 0 (3 11)		Blue
BN12080		80 0 (3 15)		
BN12081/84		82 0 (3 23)		Orange
BN12084/87		85 0 (3 35)		Purple
BN12085		85 0 (3 35)		
BN12087/90		88 0 (3 47)		Yellow
BN12090		90 0 (3 54)		
BN12090/93		91 0 (3 58)		Green
BN12093/96		94 0 (3 70)		Pink
BN12095		95 0 (3 74)		
BN12096/100		97 0 (3 82)		Brown
BN120100/103		101 0 (3 98)		Gold

PIN (see MIL-P-24705)

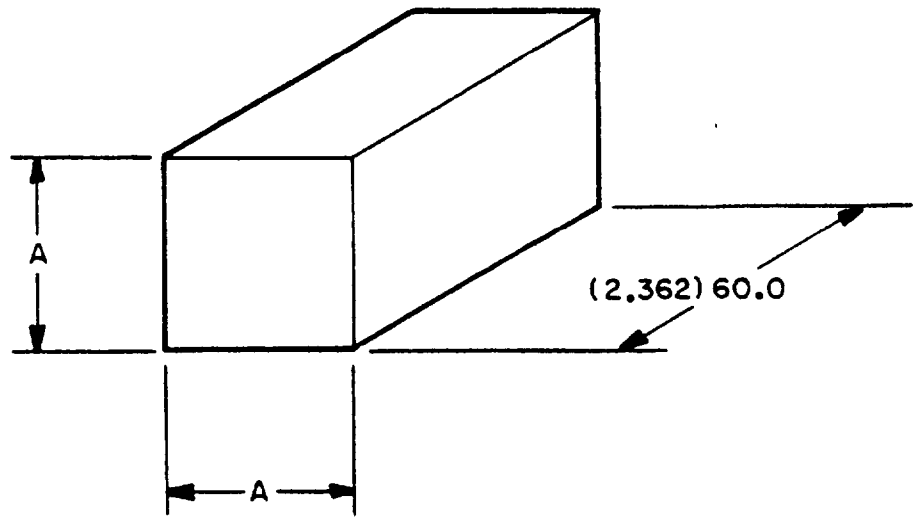
The item designation for BN120** square-faced cable insert blocks shall consist of five digits. The first three digits indicate the exterior width of the cable insert block in mm; the last two indicate the passage diameter of cable insert block in mm. The item designation for BN120**/** square-faced cable insert blocks with "COLOR DOTS" shall consist of seven to nine digits and a / after the fifth or sixth digit The first three digits indicate the exterior width of the cable insert block in mm, the two or three digits before the / indicate the minimum diameter in mm and the last two or three after the / indicate the maximum diameter in mm

EXAMPLE



NOTE: For BN80 and above, two individual blocks may be molded as one piece and slit on one side for convenience in assembly

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FIGURE 11. *Blanking insert block*

TABLE IV *Blanking insert block dimensions*

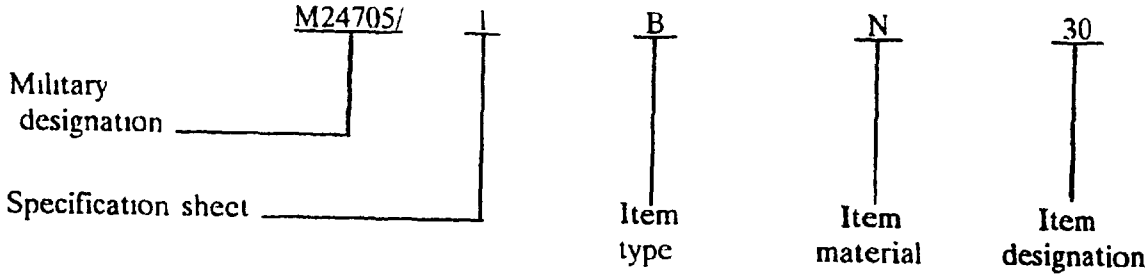
Designator	A
BN10	10 0 (0.39)
BN15	15 0 (0.59)
BN20	20 0 (0 79)
BN30	30 0 (1 18)
BN40	40.0 (1.58)
BN60	60 0 (2.36)
BN80	80 0 (3 15)
BN90	90 0 (3 54)
BN120	120 0 (4 72)

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PIN (see MIL-P-24705)

The item designation for blanking insert blocks shall consist of two or three digits that indicate the exterior width of the blanking insert block in mm

EXAMPLE



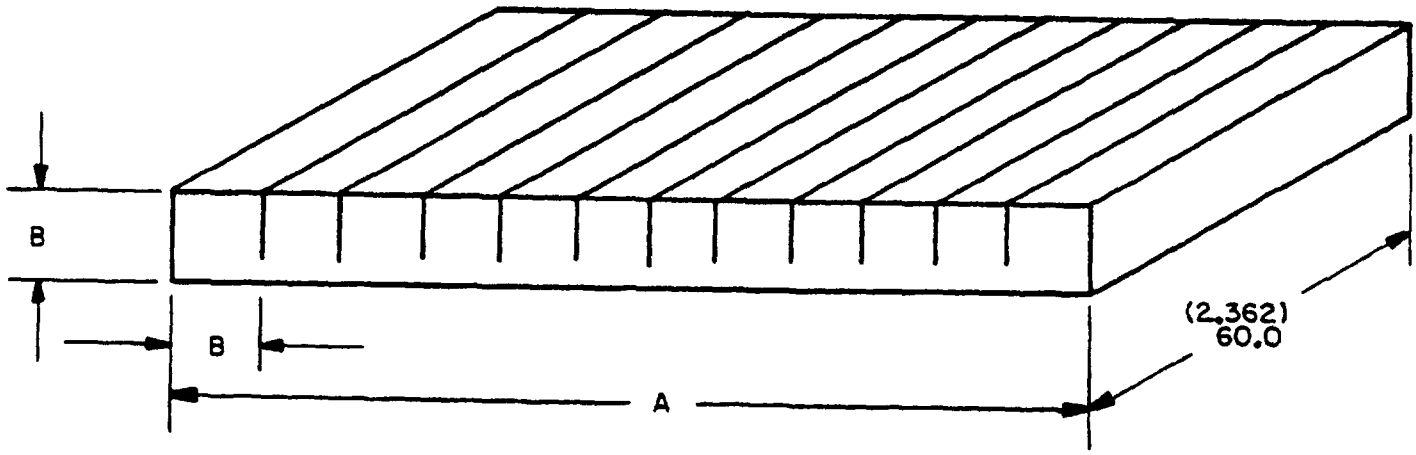


FIGURE 12 *Filler insert.*

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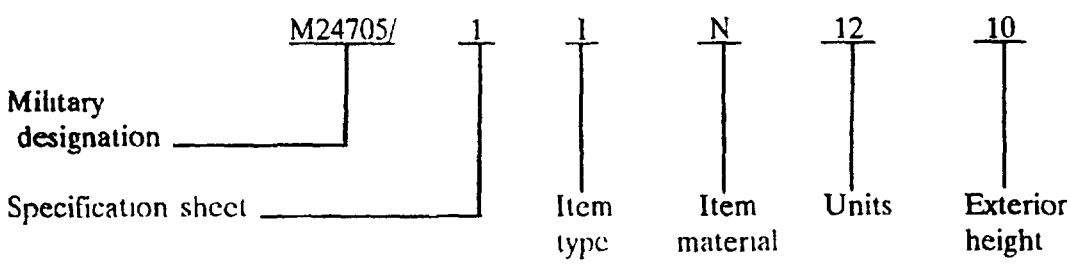
TABLE V *Filler insert dimensions*

Designator	A	B
IN0005	120 0 (4 72)	5 0 (0 20)
IN0010	120 0 (4 72)	10 0 (0 39)
IN1210	120 0 (4 72)	10 0 (0 39)
IN2405	120 0 (4 72)	5 0 (0 20)

PIN (see MIL-P-24705)

The item designation for filler inserts shall consist of four digits The first two digits indicate the quantity of filler insert units provided in the filler insert piece, the last two indicate the exterior width dimension of each filler insert unit in mm IN0005 and IN0010 are solid filler insert pieces thus no insert units are provided

EXAMPLE



Item designation

REQUIREMENTS

Nonmagnetic.	When specified (see MIL-P-24705)
Durometer·	65 to 87
Compression·	2 percent (maximum)
Expansion·	1 percent (maximum)
Sample size:	One BN60
Fluid immersion:	Plus or minus 5 percent change in durometer
Sample size:	Three BN40
IR signature:	Informational
Sample size·	One BN60

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Identification marking. Identification marking shall be required on 30 mm exterior width cable insert blocks and larger Passage diameter shall not be required on blanking insert blocks The block size number is not required for 15 mm or 20 mm blocks nor for blocks with large size passage diameters where marking space is not available On filler inserts, item designation and manufacturers name or logo shall be on the inserts, other markings are not required Optional markings on cable insert blocks shall consist of color code dots for matching insert block halves of a selected module size, cable diameter range in inches and manufacturers catalog numbers.

Rectangular frame multiple cable penetrator test specimen (see figure 13).

The components shall be as specified below Each component listed shall be required for each test specimen

Frame*	1 - F*6 (See table I)
End packing assembly	1 - E*1 (See Figure 3, 4, or 5)
Compression plate	1 - C* (See Figure 6 or 7)
Stay plate:	3 - S*2 (See Figure 8 or 9)
Two-piece square-faced cable insert blocks	2 - BN20 Insert block passage
	1 - BN30 diameter to suit
	4 - BN40 specified test cables
Blanking insert blocks	1 - BN9053/56
	3 - BN20
	2 - BN30
	1 - BN40
Filler inserts*	1 - IN0005
Cables*	LSDSGA-4, LSDSGU-4
	LSFSGA-4, LSFSGU-4
	LSMSCA-24, LSMSCU-24
	LSMSCA-30, LSMSCU-30
	LSMDU-6, LSTSGA-400
Lubricant:	C-T-91

* Represents item material See qualification table for selection.

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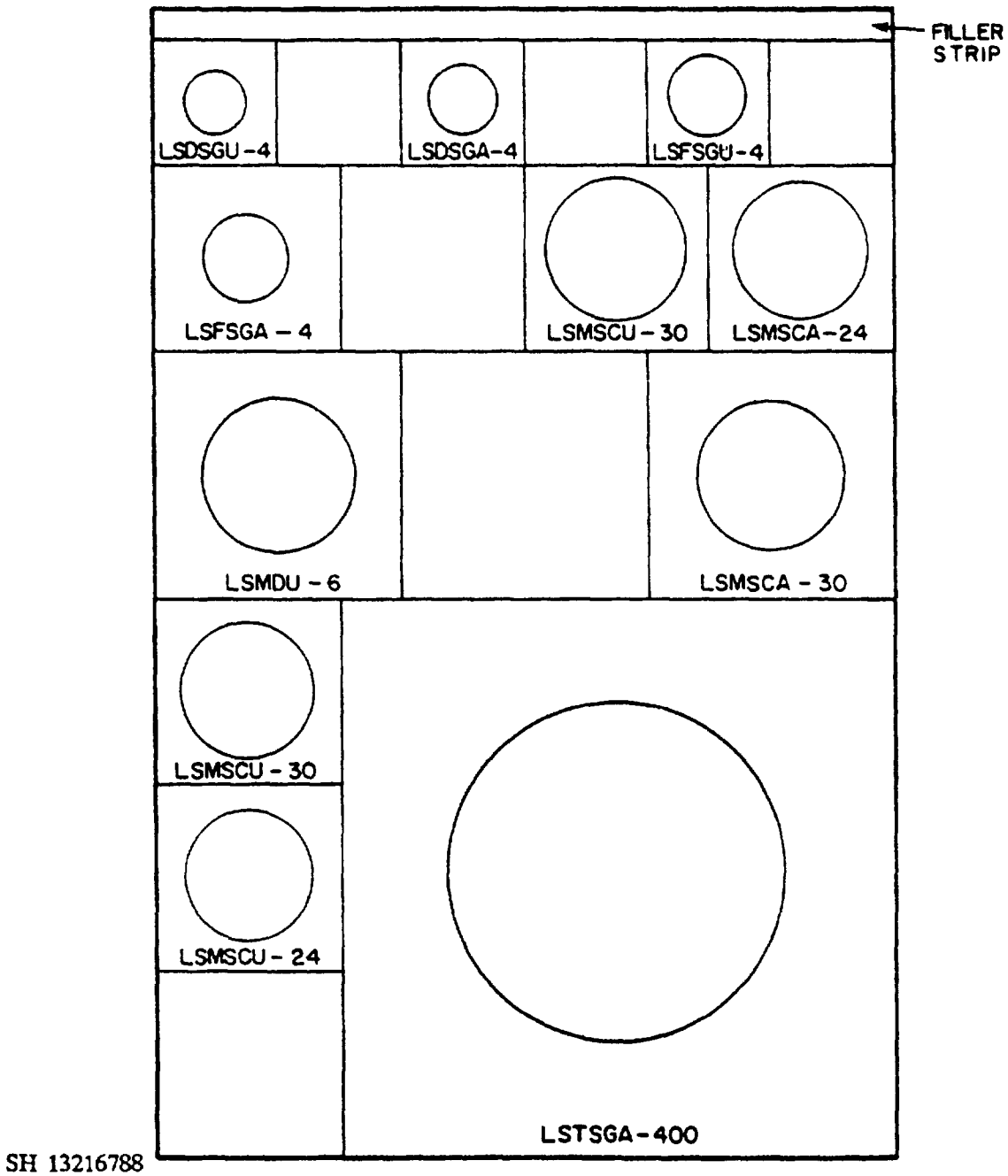


FIGURE 13 *Rectangular frame multiple cable penetrator test specimen - cable configuration.*

Preparing activity:
Navy - SH
(Project 5975-N079-02)

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